

MILESTONE 1: TECHNICAL FOUNDATION

Detection-Aware Resource Allocation for Cognitive Radio-Enabled ISAC under Compound-Gaussian Clutter

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1. Finalized Research Problem

The baseline cognitive-radio-enabled integrated sensing and communication (CR-ISAC) relay system maximizes the sum achievable rate (SAR) of secondary users while protecting the primary sensing task through a minimum echo-SINR constraint. Echo SINR measures average output signal quality, but it is not a target decision. Under heavy-tailed clutter, two systems with the same echo SINR can have different tail probabilities and therefore different probabilities of detection and false alarm.

The research question is:

How should secondary-user powers and collaborative sensing/relay beamformers be jointly designed to maximize secondary SAR while guaranteeing an explicit target-detection probability and false-alarm limit under compound-Gaussian clutter?

The intended contribution is a detector-aware optimization formulation, an analytical reformulation of its statistical constraints, and a convergent solution framework. Simulation will verify the probability expressions, feasibility, and algorithmic behaviour; simulation alone is not the research contribution.

2. System and Observation Model

The underlay network consists of an N_t -transmit/ N_r -receive dual-function base station (DBS), M single-antenna secondary transmitter–receiver (ST–SR) pairs, and K point targets. Direct ST–SR links are blocked, and the DBS acts as a two-slot amplify-and-forward relay. Channels and target responses remain constant over one coherent processing interval (CPI).

In slot 1, the DBS transmits the known sensing waveform $s[\ell]$ through $\mathbf{v} \in \mathbb{C}^{N_t}$, while ST m sends $\alpha_m x_m[\ell]$, where $p_m = |\alpha_m|^2$. For $\ell \in \{1, \dots, L\}$,

$$\mathbf{y}[\ell] = \sum_{k=1}^K \delta_k \mathbf{A}_k \mathbf{v} s[\ell] + \sum_{m=1}^M \mathbf{h}_m \alpha_m x_m[\ell] + \mathbf{c}[\ell] + \mathbf{n}[\ell], \quad (1)$$

where $\delta_k \in \{0, 1\}$ is the state of target k and $\mathbf{A}_k = \beta_k \mathbf{b}(\theta_k) \mathbf{a}^T(\theta_k)$ is its response. Receiver noise follows $\mathbf{n}[\ell] \sim \mathcal{CN}(\mathbf{0}, \sigma_n^2 \mathbf{I})$.

2.1 Compound-Gaussian clutter

Clutter is explicitly separated from receiver noise and modelled as

$$\mathbf{c}_k[\ell] = \sqrt{\tau_k} \mathbf{R}_{c,k}^{1/2} \mathbf{q}[\ell], \quad (2)$$

$$\mathbf{q}[\ell] \sim \mathcal{CN}(\mathbf{0}, \mathbf{I}), \quad \tau_k \sim \text{Gamma}(a_k, b_k). \quad (3)$$

The texture τ_k is constant within a CPI and changes independently between CPIs. This produces heavy-tailed clutter while retaining a conditionally Gaussian observation model. A normalized initial case uses $a_k b_k = 1$.

2.2 Matched-filter detector

After cancellation of known components and spatial filtering by $\mathbf{u}_k$, the coherent matched-filter output is

$$z_k = \frac{1}{L} \sum_{\ell=1}^L s^*[\ell] \mathbf{u}_k^H \mathbf{y}_k[\ell], \quad \frac{1}{L} \sum_{\ell=1}^L |s[\ell]|^2 = 1. \quad (4)$$

Conditioned on texture, the binary model is

$$\mathcal{H}_{0,k} : z_k = w_k, \quad \mathcal{H}_{1,k} : z_k = \mu_k + w_k, \quad (5)$$

$$\mu_k = \mathbf{u}_k^H \mathbf{A}_k \mathbf{v}, \quad w_k | \tau_k \sim \mathcal{CN}\left(0, \frac{\sigma_{0,k}^2(\tau_k, \mathbf{p})}{L}\right), \quad (6)$$

where

$$\sigma_{0,k}^2(\tau_k, \mathbf{p}) = \mathbf{u}_k^H [\mathbf{R}_{I,k}(\mathbf{p}) + \tau_k \mathbf{R}_{c,k} + \sigma_n^2 \mathbf{I}] \mathbf{u}_k. \quad (7)$$

$\mathbf{R}_{I,k}(\mathbf{p})$ contains secondary-user interference and other-target leakage; hence CR power allocation is directly coupled to sensing reliability.

The detector decides $\mathcal{H}_{1,k}$ when $T_k = |z_k|^2 > \lambda_k$. Its texture-averaged performance is

$$P_{FA,k} = \mathbb{E}_{\tau_k} \left[\exp\left(-\frac{L\lambda_k}{\sigma_{0,k}^2(\tau_k, \mathbf{p})}\right) \right], \quad (8)$$

$$P_{D,k} = \mathbb{E}_{\tau_k} \left[Q_1\left(\sqrt{\frac{2L|\mu_k|^2}{\sigma_{0,k}^2(\tau_k, \mathbf{p})}}, \sqrt{\frac{2L\lambda_k}{\sigma_{0,k}^2(\tau_k, \mathbf{p})}}\right) \right]. \quad (9)$$

Equations (8)–(9) are the mathematical bridge between powers/beamformers and the physical target decision. For fixed design variables, the smallest λ_k satisfying $P_{FA,k} \leq \bar{P}_{FA,k}$ is selected because $P_{D,k}$ decreases monotonically with the threshold.

3. Focused Literature Review

Hu *et al.* [1] provide the closest architecture and form the base paper. They maximize secondary SAR by jointly optimizing ST powers, sensing transmit/receive beamformers, and relay beamforming under power and echo-SINR constraints. Perfect- and imperfect-CSI cases are addressed using inexact block-coordinate and path-following methods. However, clutter is absorbed into aggregate noise and sensing is protected only through echo SINR; no detector, threshold, P_D , or P_{FA} is formulated.

Xu *et al.* [2] jointly design spectrum-sensing time, SBS beamforming, and RIS coefficients for cognitive ISAC. Their spectrum sensing determines primary-spectrum occupancy and supports radio-environment mapping; it is not the same as detecting a physical target from its echo in clutter.

Behdad *et al.* [3] derive a maximum-a-posteriori ratio-test detector for multi-static cell-free ISAC and connect target detection to power allocation. Their results evaluate P_D at selected P_{FA} values, but the network is not an underlay CR relay system.

Luo *et al.* [4] jointly design communication and sensing signals in a cluttered environment. The study demonstrates why structured clutter must not be ignored, but it does not impose detector-level reliability constraints in the CR relay

problem.

Choi *et al.* [5] develop robust multi-user/multi-target ISAC beamforming under channel uncertainty. Their uncertainty-set methodology is relevant to a later robust extension, but compound-Gaussian target detection and underlay CR relaying are outside their scope.

Zhang *et al.* [6] explicitly analyse ISAC target detection under realistic compound-Gaussian clutter. Their work supplies the statistical foundation for threshold, P_D , and P_{FA} calculations, but does not jointly allocate CR secondary powers and collaborative relay beamformers.

4. SOTA Comparison and Gap

Table 1. Focused comparison with the closest existing work. Target detection means detection of a physical target from its echo, not spectrum-occupancy sensing.

Work	CR	Relay	Joint power/BF	Target P_D/P_{FA}	Clutter	Compound Gaussian	Limitation relative to this project
Hu <i>et al.</i> [1]	Yes	Yes	Yes	No	Limited	No	Closest architecture; sensing is guaranteed only by echo SINR.
Xu <i>et al.</i> [2]	Yes	No	Yes	No	No	No	Spectrum-occupancy sensing and RIS design, not cluttered target detection.
Behdad <i>et al.</i> [3]	No	No	Power	Yes	Yes	No	Detection-aware cell-free ISAC without underlay CR spectrum sharing.
Luo <i>et al.</i> [4]	No	No	Yes	No	Yes	No	Clutter suppression without detector-reliability constraints for CR users.
Choi <i>et al.</i> [5]	No	No	Yes	No	No	No	CSI-robust beamforming without statistical clutter-aware detection.
Zhang <i>et al.</i> [6]	No	No	Input design	Yes	Yes	Yes	Detector model without joint CR power and relay beamforming optimization.
Proposed framework	Yes	Yes	Yes	Yes	Yes	Yes	Maximizes secondary SAR subject to detector-level reliability and power limits.

The component ideas are established individually, but the focused search did not identify a work that integrates them in one formulation. The research gap is the joint optimization of ST powers, sensing transmit/receive beamformers, and collaborative relay beamforming in an underlay CR-ISAC network subject to explicit per-target detection and false-alarm requirements under compound-Gaussian clutter. The novelty claim is therefore based on this specific integration, not on CR, ISAC, clutter modelling, or P_D/P_{FA} individually.

5. Proposed Optimization Framework

In slot 2, the DBS applies $\mathbf{W} \in \mathbb{C}^{N_t \times N_r}$ and relays $\mathbf{x}_r = \mathbf{W}\mathbf{y}$. Let $\gamma_m(\mathbf{p}, \mathbf{W}, \mathbf{v})$ be the SINR at secondary receiver m , including relayed target echoes, multi-user interference, clutter, and receiver noise. The rate is $R_m = \frac{1}{2} \log_2(1 + \gamma_m)$, where 1/2 accounts for two-slot relaying. The nominal problem is

$$\underset{\mathbf{p}, \mathbf{W}, \mathbf{v}, \{\mathbf{u}_k\}}{\text{maximize}} \quad \sum_{m=1}^M \frac{1}{2} \log_2(1 + \gamma_m) \quad (10a)$$

$$\text{subject to} \quad 0 \leq p_m \leq P_m^{\max}, \quad \forall m, \quad (10b)$$

$$P_t(\mathbf{p}, \mathbf{W}, \mathbf{v}) \leq P_{\text{DBS}}^{\max}, \quad (10c)$$

$$P_{D,k} \geq \bar{P}_{D,k}, \quad \forall k, \quad (10d)$$

$$P_{FA,k} \leq \bar{P}_{FA,k}, \quad \forall k, \quad (10e)$$

$$\|\mathbf{v}\|_2^2 \leq P_s^{\max}, \quad \|\mathbf{u}_k\|_2 = 1, \quad \forall k. \quad (10f)$$

In the base two-slot architecture, target detection occurs at the DBS in slot 1; therefore P_D and P_{FA} depend directly on $(\mathbf{p}, \mathbf{v}, \mathbf{u}_k)$ and clutter statistics, not directly on $\mathbf{W}$. The relay

matrix remains a joint variable because it controls secondary rate and DBS transmit power.

Problem (10) is non-convex because the rate SINRs are fractional, probabilities (8)–(9) are nonlinear expectations, and the design variables are coupled. The planned analytical solution is:

1. derive an equivalent or provably safe deterministic approximation of the P_D/P_{FA} constraints using numerical quadrature and monotonic bounds;
2. update $\mathbf{u}_k$ using a clutter-aware generalized MVDR direction;
3. calibrate λ_k by bisection to meet $\bar{P}_{FA,k}$;
4. update $\mathbf{p}$ and $\mathbf{v}$ using WMMSE and successive convex approximation (SCA) with a feasibility-preserving inner approximation;
5. update $\mathbf{W}$ using SCA or semidefinite relaxation; and
6. repeat until relative SAR improvement is below ϵ .

Each block will be constructed to preserve feasibility and not decrease its surrogate objective. Since SAR is bounded under finite power budgets, the resulting sequence is monotonic and bounded. The intended theoretical claim is convergence to a stationary/KKT point of the approximated problem, not global optimality.

6. Assumptions and Scope

Core assumptions.

- Underlay spectrum sharing, two-slot AF relaying, primary sensing priority, and no usable direct ST–SR link.

- Quasi-static flat-fading channels during one CPI; perfect instantaneous CSI in the nominal problem.
- Point targets with known nominal θ_k and β_k ; target state and clutter texture remain constant within one CPI.
- Known, unit-energy sensing waveform; mutually independent $x_m[\ell] \sim \mathcal{CN}(0, 1)$, noise, clutter, and channels.
- Known $(a_k, b_k, \mathbf{R}_{c,k})$ in the core model, plus timing/frequency synchronization and sufficient DBS self-interference cancellation.

Included in the core study: derivation and verification of P_D/P_{FA} ; threshold selection; deterministic constraint reformulation; joint $\mathbf{p}, \mathbf{v}, \mathbf{u}_k, \mathbf{W}$ design; feasibility, convergence, and complexity analysis; and comparison with the base echo-SINR method.

Planned robust extension: bounded CSI/target-response errors and uncertain clutter covariance or texture moments, using the S-procedure and a moment/CVaR safe approximation only where mathematically justified.

Excluded at this stage: waveform-code design, multi-CPI tracking, multi-cell coordination, broad hardware impairment modelling, experimental hardware validation, and simultaneous optimization over an unspecified family of detectors.

7. Testable Claims and M1 Artefacts

The study will test whether: (i) equal echo SINR can produce different P_D under different texture parameters; (ii) the base SINR-only solution violates a specified P_D or P_{FA} requirement in heavy-tailed clutter; and (iii) the proposed method finds the highest feasible secondary SAR under detector-level reliability, although its unconstrained raw SAR may be lower.

Initial MATLAB artefacts define the system configuration, generate channels and target responses, generate Gamma-textured compound-Gaussian clutter, calibrate the detector threshold empirically, and report P_D , P_{FA} , and missed detection for each target. The project repository is github.com/pratimaprasad-dot/Pratima-Prasad.

8. Evaluation Evidence

The evaluation will report ROC curves; P_D and P_{FA} versus clutter shape, clutter power, and snapshot count; SAR versus $\bar{P}_D$; feasibility probability; convergence; runtime; and the rate cost of reliable sensing. All methods will use common channel, target, power, and clutter realizations. The principal scientific outcome will identify the operating conditions under which the

base echo-SINR constraint is adequate and those under which explicit detection-aware constraints are necessary.

9. Project Timeline

The work is planned for **25 working days**. The baseline will be verified before introducing the new detector and optimization constraints.

Period	Activity	Planned output
Days 1–10	Reproduce baseline	Reconstruct the perfect-CSI system, implement the original power/beamforming procedure, and reproduce the main SAR and sensing-SINR trends, including the Figure 3 configuration.
Days 11–25	Develop new model	Derive the compound-Gaussian detector, calibrate the P_{FA} threshold, reformulate the P_D/P_{FA} constraints, construct the joint alternating optimizer, and analyse feasibility and convergence.

References

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